



=== Harwell RSX Characterization Report ===

Each resistor undergoes characterization at multiple temperature points to develop a precise model for calculating resistance across its operating range. Our modern resistors use a proprietary metal foil design that depending on the unique assembly of each resistor, we may characterize it via a polynomial or third-degree fit.

=== Instrument Summary ===

Date: September 15, 2026

Model: Harwell RS-7

Value: 25 Ω

S/N: RS70305

Public Report Reference: 6AAF040C

=== Calibration Data ===

Calibrated Resistance: 25.003691 Ω at 23°C

Expanded Uncertainty of Measurement, k=2: 0.8 PPM

Measurement Current: 1 mA

=== Temperature Characterization Data ===

α : -2.864E-07 $\Omega/\Omega/^\circ\text{C}$ (-0.286 ppm/ $^\circ\text{C}$)

β : -1.948E-07 $\Omega/\Omega/^\circ\text{C}^2$ (-0.195 ppm/ $^\circ\text{C}^2$)

Linear TCR (18-28°C): -0.31 ppm/ $^\circ\text{C}$

To calculate the resistor value at a temperature other than 23°C, use the following model where T is the desired temperature in $^\circ\text{C}$:

$$R(T) = R_{(23)}[1 + \alpha(T - 23) + \beta(T - 23)^2]$$

Note: This model may not be accurate outside the characterized temperature range.

Temperature ($^\circ\text{C}$)	Temperature ($^\circ\text{F}$)	Resistance (Ω)	Deviation (ppm)
18.0	64.4	25.003605	-3.44
18.5	65.3	25.003625	-2.66
19.0	66.2	25.003642	-1.97
19.5	67.1	25.003656	-1.38
20.0	68.0	25.003669	-0.89
20.5	68.9	25.003678	-0.50
21.0	69.8	25.003686	-0.21
21.5	70.7	25.003691	-0.01
22.0	71.6	25.003693	0.09
22.5	72.5	25.003693	0.09
23.0	73.4	25.003691	0.00
23.5	74.3	25.003686	-0.19
24.0	75.2	25.003679	-0.48
24.5	76.1	25.003669	-0.87
25.0	77.0	25.003657	-1.35
25.5	77.9	25.003643	-1.93
26.0	78.8	25.003626	-2.61
26.5	79.7	25.003606	-3.39
27.0	80.6	25.003584	-4.26
27.5	81.5	25.003560	-5.23
28.0	82.4	25.003533	-6.30

Harwell Resistor TCR
Model: RS-7 | Nominal Value: 25 Ω | Serial Number: RS70305

